Enhancing the Shopping Experience through User-Centered Design: Insights and Recommendations from the Development of SSN"

Chiyu Wang
for Submission
Kingston, Canada
19cw33@queensu.ca

Allen Zhang
for Submission
Kingston, Canada
19xz64@queensu.ca

Weihan Xie
for Submission
Kingston, Canada
20wx10@queensu.ca

Elaine Wu
for Submission
Kingston, Canada
19yw78@queensu.ca

Ziguang LIU
for Submission
Kingston, Canada
20zl104@queensu.ca

ABSTRACT

The development of the "Super Sales" (SSN) platform aimed to provide a comprehensive solution for users to find the best deals and discounts for their online and offline shopping. Through user testing and evaluation, the project team was able to gain valuable insights into the user experience and identify areas for improvement. The findings showed that the app was well received by most participants, but there were also areas for improvement, such as the need for a simpler design for younger users and the addition of accessibility features for visually impaired and elderly users. The evaluation processes were critical in shaping the design of the app and ensuring that it meets the needs and expectations of users. The project highlights the importance of user-centered design principles and accessibility in creating a user-friendly and accessible platform. The insights and recommendations from the development of SSN could be valuable for other UI/UX designers and researchers in the field, as they demonstrate the importance of considering user needs and experiences in the design and development of novel technology.

INTRODUCTION

Our team is planning to launch a mobile application that collects information on food discounts within three months. The name of this application is Super Sales, correspondingly abbreviated as SSN, and it is developed for OS and Android platforms. The inspiration that led to this idea is closely related to our lives, we often read online or chat with friends about the negative emotions of disappointment or regret for missing out on offline or online food sales. We explored this phenomenon more deeply, conducting a 9-question questionnaire and interviews. According to the 72 survey data, 86% of the people were not informed of the food discount in time. There are three specific reasons for this. The first is the fragmentation of information, as many online shopping sites and offline grocery store sites (Amazon, Walmart, Costco, Metro, Food Basics, etc.) already exist in the network, and the discounted food/prices will vary from site to site, making it inconvenient for users to find information. The second reason is the popularity of food. Based on our research on each website, we found that most websites will first push more advanced/technological/expensive items (monitors, coffee machines, etc.). At the same time, food is in a relatively backward, hard-to-find position. The third reason is that some offline supermarkets do not have websites, such as Chinese and Korean markets. The date the customer goes to the supermarket may miss the day when the food is on sale. In addition, 67% of people in this survey desire to use software that organizes news about food discounts. Reflecting on the feedback received, we made the decision to define Super Sales as an application that provides a comfortable, efficient, easy-to-use, and aesthetically pleasing environment for users to collect information on discounted food products. The application is divided into three main parts, the fixed crawler code is used to find sales activities every day, publish specific information about the events, and users can participate in contributing to promoting offline sales activities in the application at any time.

RELATED WORK

In terms of its relevance to the interaction design community, “Super Sales”(SSN) has made significant contributions to the development of user-friendly and efficient delivery and pickup systems. And integrate online and offline sales information to provide users with one-stop shopping information and shopping channels. The SSN app and website should be designed to be simple and intuitive, making it easy for users to order groceries [1] and track their delivery and make and obtain the local market’s sales’ information immediately, and help users make the best shopping decisions for them.

In terms of its relevance to the broader HCI community, SSN will help to advance research in areas such as e-commerce, mobile computing, and location-based services. The use of data and machine learning algorithms to personalize the shopping experience for customers has also been a topic of interest for researchers in HCI.[3]

Furthermore, our team plans to apply deep learning and big data technologies to predict prices from local markets and online shopping websites for users in the coming days. If this technology can be implemented, we believe it will disrupt the current users’ shopping habits.[4]

DESIGN CONCEPT

In today's society, online and offline shopping has become a daily activity for people. However, shopping also poses many challenges, not only for consumers but also for merchants. For example, consumers often miss out on deals on items such as perishable foods, especially those with a short shelf life. This leads to lost sales for merchants and missed opportunities for consumers to purchase essential items at lower prices. SSN solves this problem by unifying discount information from online and offline sales channels to provide users with a convenient shopping experience. In addition, SSN has built a communication bridge between merchants and consumers, which enables merchants to reduce advertising expenses on this platform, and users can purchase the products they need at a more favorable price. SSN aims to create a win-win situation for both merchants and consumers.

USER CHARACTERIZATION

SSN aims to provide a comprehensive solution for people of all ages to find the best deals and discounts for their online and offline shopping. The target user group of SSN includes both men and women, and they can come from all different cultural backgrounds, but one thing they have in common is that they are budget-conscious, convenience-oriented, and want to get the best deals and discounts when they shop. The software's user demographic can be divided into budget-conscious shoppers, convenience-focused shoppers, impulse buyers, frequent shoppers and new shoppers, and the software provides personalized service to users by targeting skilled technology.

For people in the younger age group (e.g., under 25), with low income, or just starting on their own, SSN provides guidance and advice on products and the best discounts; for users between 25 and 45, SSN can significantly improve their daily lives and help them make spending decisions that save them both time and money; for older users, SSN provides easy-to-understand instructions and shopping processes.

To make SSN accessible to as many people as possible, our UI design respects the user's established operating habits, and all commands in the software are done with finger clicks and swipes, allowing users to use the software immediately without having to invest in learning how to use it. In addition, SSN provides a shopping experience for some people with physical disabilities, for illiterate users, we used the most commonly used words and easy-to-understand graphics to communicate the user's needs; for people with low vision, we used a large number of images in the design of the software to convey information to the user, rather than conveying information to the user in textual descriptions., etc. Moreover, taking into account the user's eating habits, the user can select their taboos when they first log in to block out information they do not need. Furthermore, the software is Designed to work seamlessly across mobile devices, online shopping sites, social media platforms and in-store shopping environments to ensure user-friendliness and accessibility. Overall, SSN can provide an excellent and enjoyable shopping experience to a wide variety of user groups.

USAGE SCENARIOS

Online Shopping Case:

With limited time, a busy workaholic named Jane needed a quick and convenient source of energy and nutrition. Before using SSN, she had to go downstairs to the supermarket every time to buy instant food, which she felt was a hassle and would interrupt her work, thus affecting her productivity. At a colleague's party, she recommended SSN to Jane, who turned to SSN for its ease of use and real-time delivery tracking. When she is using SSN, Jane has filtered out the store with the highest cost-effectiveness and placed an online order with this store. While waiting for the goods to be delivered, she can track the ordered items in real-time. She was so happy that she can compare prices, read product reviews, and make informed purchasing decisions through the app. Using SSN was more effortless than ever for Jane, and the shopping time was short, so she did not have to worry about interrupting her work. Finally, her functional drink was delivered to her doorstep, which made her feel so happy that she decided to use SSN instead of going to the supermarket in the future.

Offline Shopping Case:

Rocky, a student at Queen's University, found himself facing a busy day with two assignments due two days after returning from vacation. Before using SSN, Rocky always went to the supermarket downstairs to buy a can of Monster to keep himself energized for his evening work, but he was so confused about choosing which stores to shop around his house. During a club event, Rocky learned about SSN and this time, he chose to use it, finding it very convenient to compare the prices of products from three stores in the area. The platform's mastery of individual product prices allowed Rocky to quickly determine that the nearest Shoppers were the most expensive option, while the University store offered the best prices. With the ability to plan his route and get the best cost savings for himself, Rocky chose to buy his Monster on campus, and he felt very pleased about this shopping experience because it saved him money on other items he liked. Rocky compared prices on SSN before planning his route in every subsequent purchase.

The case of a Visually Impaired person:

A visually impaired person named Sarah who uses Instacart's voice interaction feature to purchase fruit. As someone who has difficulty navigating the aisles of a grocery store, Sarah is excited to have the option to make her purchases online and maintain her independence. When using Instacart, Sarah only needs to open the app through her phone's voice assistant. Then she can continue to select stores and products using voice commands, and the voice assistant will provide real-time information on the selected product's details and its prices, as well as any discount information available at other stores. After placing her order, the voice assistant will also track Sarah's ordered products in real-time and broadcast the order status to her, allowing her to stay informed at all times. Sarah is so pleased with this convienent shopping experience. In the end, her fruits are delivered to her door, allowing her to enjoy the convenience and confidence that comes with using Instacart's accessible technology.

The case for the people who have the prohibition:

Eric believes in Islam, so he does not eat pork. Before using SSN, he is unfortunate that he sees some information about pork on his online shop app. However, an advertisement showed him a new shop app called SNN. So, when he first logged into SSN, he chose not to receive pork-related product pushes. After that, he never received any pork-related products pushed to him, but other foods. He is very pleased about this function and has begun to shop by SSN instead of the previous online shop app.

The case for an elderly:

The elderly Mrs. Johnson, a senior citizen with limited mobility, is perplexed whenever he shops online on other online shop apps because it is hard for the elderly to do some operations. He knew about Instacart from his child's recommendation. When he first uses Instacart to purchase eggs with ease. The app's user-friendly GUI design, including large and clear text, simple navigation, high-quality product images, clear product information, and an easy checkout process, empowers her to manage her grocery shopping without relying on help from others. When he is using Instacart, he could easily find the goods by searching it on the app, and all the prices and discount information about it will be shown, and then he could choose the most preferable one and order it directly without too much diffcult operation. He is delighted with his shopping experience on Instacart. Also, Mrs. Johnson receives real-time delivery updates through text notifications and enjoys the convenience and independence offered by Instacart. Ultimately, he likes to shop by Instacart instead of the previous online shop app.

DESIGN

According to the first draft of the design, we used the logo of each store's website instead of the textual store name in the first draft. We also used different background colors to distinguish the online shopping from the offline shopping sub-interface, instead of simply using lines to divide the stores. We chose to use rounded corners for the store icons to keep the consistency with the interface of the phone itself, so as to give users a better feeling and more comfortable visual effect. Also, we combine the store's logo and name with rounded-corner frames to avoid ambiguity. Different colors are used to mark the difference between online and offline stores. User can click on the see more button to view more stores.

Figure 1.1 Homepage

In the first draft of "Store List" and "items list", we used hand-drawn product icons as examples to design the product display structure of "Online Store" Now, we are using the actual pictures provided by each supermarket's website to display the products, so as to keep the consistency and authenticity between the products displayed in SSN and the products on the corresponding store website, and to make these pictures more valuable for reference when users view the products. All product images have been reformatted to the same size to keep the page tidy. Like the design of Homepage, the store icons are designed with rounded corners to maintain consistency with the interface of the phone itself, which gives users a better feeling and more comfortable visual effect.

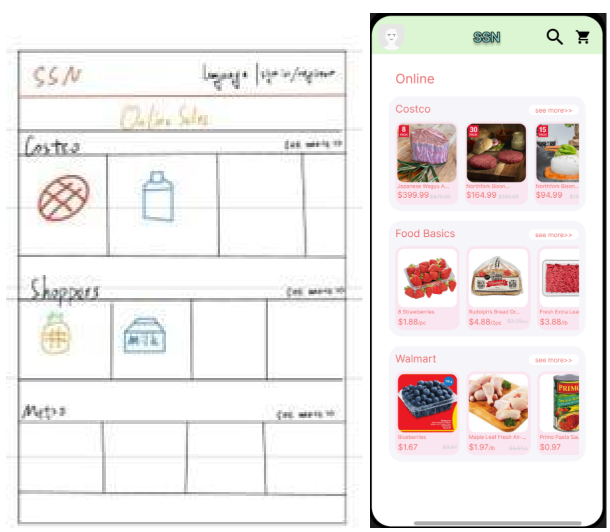

Figure 1.2 Store List

Figure 1.3 Items list

In the product details screen, we optimized the interface with more details than the first draft. We chose to give more space to the product images and more space to the product details. We chose to weaken the display of the original price and enhance the display of the discounted price to help users better grasp the important information of the product. The "ADD TO CART" button is designed as a rounded button with a pink background, making it more prominent among the dark blue-gray product descriptions. Additionally, the button contains a shopping cart icon to accommodate the usage habits of multiple user groups, including the elderly. And display similar products below the products to give users more choices.

Figure 2.1 Item information

Furthermore, at the beginning of the design, SSN was not designed to have online shopping functionality, but after a group discussion, we decided that online shopping was necessary for SSN, so we added this functionality later on and designed the UI interface for online shopping.

HI-FI DEVELOPMENT

Based on the results of thorough user testing, the high-fidelity prototype of the Super Sale online shopping application is undergoing refinement and improvement. Our design and development team is working diligently to optimize the user experience and make sure that the application is both user-friendly and efficient. The changes being made to the prototype include updates to the functionality, such as improving the search capabilities and adding new payment options, as well as enhancements to the overall user experience, such as simplifying the navigation and making the interface more visually appealing. Our ultimate goal is to create a final product that exceeds the expectations of our users and provides a seamless shopping experience.

After a group discussion, we decided to use figma to create the interaction design of the software. We also used dopelycolors for the color design of the software interface. Before deciding on the application color scheme, we divided the application into different parts that require different colors: the background color of the application, online stores, offline stores, their respective text requirements, and other text. We then selected the appropriate color scheme accordingly. Since the interface we design is for listing grocery products and their discount information, we chose light green as the background color of the application and used a normal green for the logo. The application focuses more on providing online shopping items and discount information, so we chose bright pink as the background color and red to emphasize the text. For the offline stores, we chose light blue and darker purple. To make the interface look simple and neat, and to highlight important information such as product names, prices, and discounts, we designated long text including descriptions, and other minor information as a dark blue-gray color.

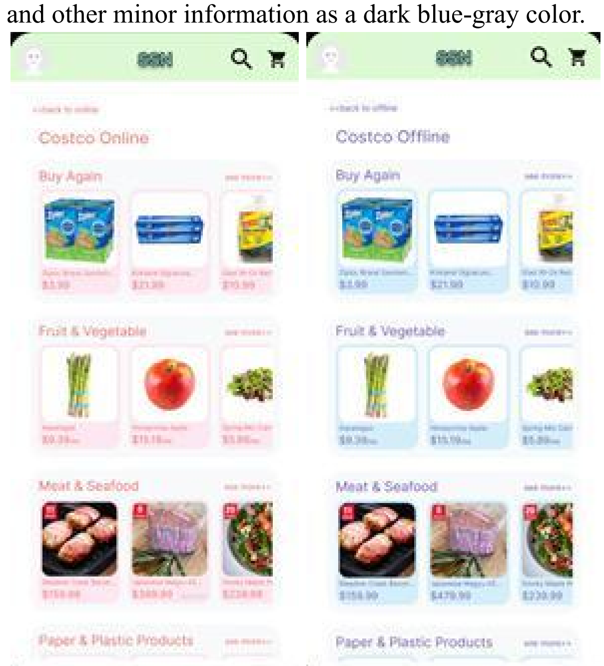

Figure 3.1(Online and Offline Design)

Testing:

We divided the test into three parts. The first part is a test of the SSN online shopping function, using the demo function of figma to test the smooth operation of the 3 stages of the online function: choosing a store, selecting items and buying items. The first part is the test of the offline shopping function, which is to test the smooth operation of the three stages of selecting a store, selecting an item, and viewing the item information. Finally, we test whether users can jump directly to the offline shopping interface when viewing online products.

Evolution:

SSN was originally designed as a website to help users look up discounts at various supermarkets; it did not have the ability to help users with online shopping. And afterwards, considering that SSN was originally designed to enable users to purchase items at a better price and to save the time they spend while shopping, we asked the question "Does it really save users' time to use an application to download and use another software for online shopping?" After some discussion, we decided to collect discounts from various supermarkets and give the software SSN the ability to shop online, so that users would not need to jump to other websites to make purchases after deciding on the items they wanted to buy. After several discussions, we came up with this shopping software that combines online shopping and offline discount checking.

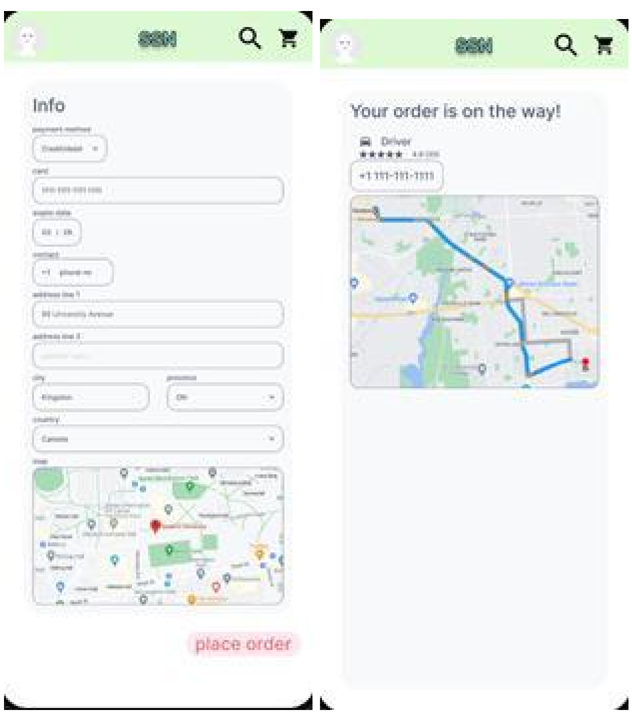

Figure 3.2 Place Order page

USER EVALUATION

Method:

We chose to let users try the operation directly through the demo function of figma, and guide them when they encounter confusion (but not give answers directly), and improve the interaction scheme of SSN by simulating and recording the process of using SSN independently for the first time, and recording the confusion encountered during this process to reduce the learning cost of users using SSN as much as possible. Because the original intention of SSN's interactive interface design is to enable consumers to operate the software with the lowest learning cost.

Based on A2's feedback, we redesigned the structure of the study based on topic classification to argue for the actual user experience and usability of SSN.

Interview Questions:

How easy was it to find what you were looking for on our application?

a. Very easy

b. Somewhat easy

c. Neutral

d. Somewhat difficult

e. Very difficult

Did you experience any technical difficulties while using our application?

a. Yes

b. No

How would you rate the overall design and layout of our application?

a. Excellent

b. Good

c. Neutral

d. Poor

e. Very poor

Were you able to complete the task you set out to do on our application?

a. Yes

b. No

How would you rate the user experience on our application compared to similar websites/applications you have used in the past?

a. Much better

b. Somewhat better

c. About the same

d. Somewhat worse

e. Much worse

What suggestions do you have for improving the user experience on our application? (Please specify)

Would you recommend our application to others?

a. Yes

b. No

How likely are you to use our application again in the future?

a. Very likely

b. Somewhat likely

c. Neutral

d. Somewhat unlikely

e. Very unlikely

Is there anything else you would like to share about your experience with our application? (Please specify)

Data Collect:

In user research, we observe participants and record their behavior and interactions with the product and audio/video recordings (providing a detailed record of participant behavior). Surveys may also be conducted to gather additional insights into participants' experiences and opinions to obtain data. Data collected in user studies can include demographic information such as participants' age, gender, education, and occupation, as well as behavioral data such as their actions and interactions with the product, time spent on the task, and performance data such as accuracy and speed.

Example 1:

We let User A pick the daily necessities, and then decide on the purchase with the information provided by SSN. After reviewing the SSN, User A decides to purchase toothpaste through an offline purchase. This is because after checking the SSN, User A finds that the price of toothpaste purchased online is the same as the price of offline purchase, and because using online purchase, User A will have to pay shipping and service fees. And by finding the location of the store, User A thinks the store is close to his location and can walk to the store to buy toothpaste.

In this example, User A did not encounter any obstacles in the process, and he was able to use SSN to find out the price difference between online and offline sales channels and make the right decision for himself. The downside is that we think the SSN could show the store location directly below the offline purchase details, instead of letting user A look up the location using Google Map. This would save users' time and achieve the purpose of SSN.

Example 2:

User B wants to purchase meat products through SSN. When user B uses SSN to check the price of beef in both online and offline sales channels, he finds that the online price is cheaper, so he decides to buy beef online.

When User B checked the price of beef, we found that after checking the price of beef in the online channel, User B could easily jump to the beef product screen of the offline channel to check the details. However, when he decided to buy online after comparing prices, he found that he could not jump directly from the offline sales channel to the online purchase screen, but had to go back to the home page and re-enter the online purchase screen corresponding to the head of the cow. This is very inconvenient, and we need to make improvements in the follow-up.

Example 3:

User C, a user with high myopia, did not wear glasses to use SSN in order to test whether the interactive interface of SSN still works well in the case of visually impaired users. “During use, I felt that I was sometimes unable to press the exact buttons I wanted, especially the smaller ones like 'Back' and 'See More'.”

Based on our observations and the subjective feelings expressed by user C, we found that some of the buttons were too small, which led to User C often mistakenly touching them during operation, thus wasting time and affecting the experience.

Findings:

First, we found that the buttons on the software needed to be larger to make it more user-friendly and reduce the possibility of accidental touches.

Second, we need to add a map to the detailed screen for viewing products for offline shopping to show the distance between the user and the store they are viewing.

Thirdly, we need to add a button to jump to the corresponding online product in the detailed screen of offline shopping products to make it more convenient for users and save time.

Recommend:

Based on the findings, it is recommended to improve the user experience of the software by increasing the size of the buttons, adding a map to the detailed screen of offline shopping products to show the distance between the user and the store, and adding a button to jump from the detailed screen of offline shopping products to the corresponding online product. These changes will make the software more user-friendly, reduce the risk of accidental touches, improve the convenience and efficiency for users, and enhance the overall user experience.

DISCUSSION & REFLECTION

Introduction:

The aim of this project was to design and develop a mobile application that would provide users with information on product discounts available at various stores. This discussion section highlights the key insights and lessons learned from user evaluation and user testing, and the originality of the project work in light of previous relevant work. We also discuss the limitations of our project, future work, and provide a thoughtful reflection on our design process, technical reflections, and conclusion.

Key insights and lessons learned from user evaluation and user testing: During the user testing phase, we gathered feedback from a diverse set of users to understand their interactions with the application and the flow of the experience. We found that users liked the visual look of our application, and many users remembered the background Settings and rounded corner store icon of the SSN, which can lead to a much higher usage of the SSN than using other applications.

Another important insight from users' feedback is maintaining the consistency and authenticity of the displayed products and the corresponding store products, such as the product name, price, quantity, and packaging. This consistency reinforces the authenticity of the products and enhances the user experience. When users browse the product details, they should feel confident that they are getting a genuine product and that there are no surprises when they arrive at the store or receive their online order.

Limitations:

Our project has some limitations that should be taken into account. First, it depends on the availability and accuracy of discount information provided by different stores. If the store does not regularly update its discount information or provides inaccurate information, the application may not be able to provide the user with reliable up-to-date information. For example, the consistency and authenticity of the products displayed by SSN with those on the corresponding store's website may lead to frustration and negative user experience if the store does not update their product pictures or information in a timely manner. Therefore, it is important for the store to maintain accurate and up-to-date information to ensure the success of the application.

The model of SSN relies heavily on third-party retailers to deliver its services, which means that it lacks direct control over the quality measures of the products being delivered. This can lead to a situation where the desired standards of quality are not met, potentially affecting the customer experience. Additionally, customers shopping online usually have the ability to choose alternative products if the item they desire is not available. However, this option may not be available to shoppers, who may have to choose the closest substitute, potentially compromising the overall shopping experience for customers.

Future Work:

In the future, we could incorporate a feature that allows users to share information on discounts they have discovered with other users of the application. This could increase the amount of information available on discounts and improve the overall user experience.

In addition, Enhance communication between courier and store and consumer, as communication is key to getting the right order delivered to the right person at the right time. Scheduling is a key feature that allows consumers to manage delivery times according to their convenience. This will be a great advantage for the SSN.[2]

We could also consider incorporating a feature that allows users to set notifications for discounts on specific products or stores they are interested in. This would provide users with even more personalized and relevant information and enhance the value of the application.

If given the chance to start the project again in the future, we would have done more extensive user testing and prototyping to gather feedback and ensure a smoother user experience. Additionally, we can supply our own products, so that we can have direct control over the quality of our products.

Reflection on the design process:

Our design process revolves around creating a user-friendly application. We emphasize the details of the products that SSN offers and aim to provide users with a seamless experience so that they can purchase the products they want with confidence. At the beginning of the design, we did not offer similar products to users, but now SSN offers similar products under the product. We also put a lot of emphasis on the aesthetics of the app, focusing on creating a visually appealing modern design, such as the rounded corner design of the store icons, and with different background colors to distinguish online and offline shopping interfaces to attract and appeal to a wide variety of users.

Originality:

In terms of originality, SSN's design stands out from previous related work due to its use of actual product images, distinguishes online from offline shopping areas, and the decision to add online shopping functionality. These elements make the app more user-friendly and engaging, which sets it apart from other similar apps in the market.

Conclusion:

In conclusion, the development of SSN has been a valuable learning experience for our team, highlighting the importance of user-centered design principles and accessibility in creating a user-friendly and accessible platform. The evaluation processes have been critical in shaping the design of the app and ensuring that it meets the needs and expectations of users. Our work has shown the potential for using technology to improve the shopping experience for users, and there are opportunities for expansion based on the feedback from users. These findings and design recommendations could be valuable for other UI/UX designers and researchers in the field, as they demonstrate the importance of considering user needs and experiences in the design and development of novel technology.

REFERENCES

[1] "HeinOnline." HeinOnline, https://heinonline.org/HOL/LandingPage?handle=hein.journals/clr108&div=28&id=&page=. Accessed on March 3, 2023.

[2] B Strategy Insights. (2021, October 8). Instacart SWOT Analysis 2021: Key Strengths, Weaknesses, Opportunities and Threats. Retrieved March 3, 2023, from https://www.bstrategyinsights.com/instacart-swot-analysis/

[3] IncFact. (n.d.). Shopkick Inc. Redwood City, CA. Retrieved March 3, 2023, from https://incfact.com/company/shopkick-redwoodcity-ca/

[4] Kim, J., Kim, H., & Lee, J. (2017). An Empirical Study on the Applicability of Convolutional Neural Networks to Text. ACM Transactions on Asian and Low-Resource Language Information Processing, 16(3), 18:1-18:15. doi: 10.1145/3001938 from https://dl.acm.org/doi/10.1145/3001938